

Kieran Dalton

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SUMMARY

Experimental quantum physicist completing a PhD at ETH Zurich, specializing in the design, fabrication, and measurement of superconducting qubits. Experienced in taking 3D-integrated and multi-module quantum processors from initial concept, to simulation, to design tape-out, to cryogenic measurement. Scaled lab architecture from zero to 17 qubits and developed new on-chip components. Seeking to apply my hands-on background in device design and EM simulation, cleanroom fabrication, and quantum device measurement to advance fault-tolerant hardware development in industry.

EDUCATION & RESEARCH EXPERIENCE

ETH Zurich

2022 – Present

PhD, Physics – Quantum Device Lab

Zurich, Switzerland

- **3D Integration & Scaling:** Led the development of 3D-integrated superconducting qubits from inception. Developed automated design code and ran fabrication/targeting tests to implement the lab's first 3D-integrated qubit devices and multi-module devices. Scaled architecture from zero to 17 qubits while developing inter-module couplers that match the operation fidelities of established planar devices.
- **Hardware Innovation & Measurement:** Engineered novel components harnessing 3D integration, such as a wiring-efficient XYZ line, optimized via Ansys HFSS (FEM) and lumped-element models. Deployed multi-module quantum processors, taking full ownership of the experiment: from chip and PCB design, through fabrication iterations, to cryostat wiring, electronics setup, and final cryogenic measurement. Implemented 6-qubit cross-platform verification algorithms and measurement-based blind quantum computation between modules.
- **Leadership & Lab Operations:** Supervised 7 student projects spanning cleanroom fabrication, chip design, TLS tuning, and the execution of a 6-qubit experiment. Organized the acquisition of >650k CHF in fabrication equipment and managed ongoing tool maintenance.
- **Academic Management:** Served as Head Teaching Assistant for master's-level courses, taking responsibility for overall course management, teaching staff coordination, and writing exams.

University of Cambridge

2021 – 2022

MSci, Physics (Grade: First – Awarded Foundress Prize)

Cambridge, UK

- **Masters Project:** Quantum algorithms for molecular simulations (Supervised by Dr. David Arvidsson Shukur).
- Simulated ADAPT-VQE and QEB-ADAPT-VQE under noise models, comparing performance and identifying limitations for NISQ applications. Utilized CUDA, OpenFermion, and Qiskit.

University of Cambridge

2018 – 2021

BA, Physical Natural Sciences (Grade: Double First)

Cambridge, UK

- Third year results: First class, 20th in cohort. Awarded Foundation Scholarship and Foundress Prize.
- First year results: First class, 4th in cohort of 604 students. Awarded Satish Kumar Aggarwal Prize for outstanding first-year performance in Mathematics or Natural Sciences.

PUBLICATIONS

Performance Characterization of a Multi-Module Quantum Processor with Static Inter-Chip Couplers

2026

G. J. Norris, K. Dalton, D. C. Zanuz, A. Rommens, A. Flasby, M. B. Panah, J. C. Besse, and A. Wallraff
<https://doi.org/10.1140/epjqt/s40507-026-00469-z>

Resource-Efficient Cross-Platform Verification with Modular Superconducting Devices

2025

K. Dalton, J. Knörzer, F. Hoehne, Y. Song, A. Flasby, D. C. Zanuz, M. B. Panah, I. Besedin, J. C. Besse, A. Wallraff
<https://doi.org/10.1103/czph-xpzs>

Quantifying the effect of gate errors on variational quantum eigensolvers for quantum chemistry

2024

K. Dalton, C. K. Long, Y. S. Yordanov, C. G. Smith, C. H. W. Barnes, N. Mertig, and D. R. M. Arvidsson-Shukur
<https://doi.org/10.1038/s41534-024-00808-x>

Layering and subpool exploration for adaptive Variational Quantum Eigensolvers

2024

C. K. Long, K. Dalton, C. H. W. Barnes, D. R. M. Arvidsson-Shukur, and N. Mertig
<https://doi.org/10.1103/PhysRevA.109.042413>

TALKS & PRESENTATIONS

MIT Seminar	Mar 2026
<i>3D-Integrated Superconducting Quantum Devices</i>	Cambridge, MA
APS Global Physics Summit	Mar 2026
<i>Wiring-Efficient XYZ Line for 3D-Integrated Superconducting Devices</i>	Denver, CO
APS Global Physics Summit	Mar 2025
<i>Cross-Platform Verification with Modular Superconducting Devices</i>	Anaheim, CA
Quantum Cavities Workshop (Poster)	May 2024
<i>High-Fidelity Inter-Module Two-Qubit Gate on a 3D-integrated Superconducting Device</i>	Jouvenne, Canada
ETH Quantum Engineering Commission Seminar	Oct 2023, Mar 2024
<i>Performance constraints of NISQ quantum algorithms and their mitigation with 3D-integrated superconducting quantum devices</i>	Zurich, Switzerland
QIP Conference 2022, Linnaeus University	June 2022
<i>High-performance density matrix simulator for testing VQE methods using iteratively-grown ansätze</i>	Värjö, Sweden

TECHNICAL SKILLS AND SERVICES

Hardware & Cleanroom Fabrication: Photolithography, Etching, Thin-Film Deposition, Flip-Chip Bonding.
Cryogenics & Microwave Engineering: Dilution Refrigerators, RF/Microwave Test Equipment (VNAs, AWGs), Cryogenic Wiring & Filtering, Qubit Control & Readout.
Simulation & Modeling: Ansys HFSS (FEM), Lumped-Element Circuit Modeling, PCB Design, Hamiltonian Parameter Extraction, Electromagnetic (EM) Simulation, Yield & Crosstalk Analysis.
Software & Programming: Python, MATLAB, Git/Version Control, C++, CUDA, Linux/Bash, L^AT_EX.
Quantum Information & Control: Qiskit, OpenFermion, Automated Hardware Design (e.g., GDSPy, KLayout), Lab Instrument Control, Variational Quantum Algorithms (VQE).
Manuscript reviewing: Physical Review Letters